

Localization error and the speed–persistence coupling of migrating cells

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Faster migrating cells tend to move more persistently, a relation proposed to be universal across cell types. We show that centroid localization error alone produces this relation in a population of cells whose speed and persistence are independent. Error of standard deviation σ per coordinate lowers the lag-1 correlation of consecutive steps by an amount that grows as the cell’s displacement per frame shrinks. As a result, slow cells look less persistent. For a persistent random walk we give the measured step correlation and mean turning cosine in closed form, together with an exact per-cell inversion. Because σ cannot be reliably estimated from short tracks, we report a sensitivity curve: the fraction of the observed speed–persistence gap that noise of size σ would account for. We applied it, under protocols fixed before the data were analysed, to public track sets from three studies. In three in vivo lymph-node datasets, T cells and neutrophils show no positive gap in the noise-correctable measure, and the B-cell gap would be half explained by $\sigma \approx 0.3\text{--}0.4\ \mu\text{m}$. In vitro T cells on ICAM-1 and VCAM-1 show a strong coupling, but σ of 0.6 and 0.4 μm respectively, under one camera pixel, would explain half of it. Nearly immobile tracks give a conservative lower bound $\sigma \gtrsim 0.2\ \mu\text{m}$, so noise accounts for at least 12–14% of the in vitro coupling. The obvious upper bound from the same tracks is biased by selection and invalid. Long tracks change the picture. In the zebrafish T-cell data of Jerison and Quake, where the speed–turning coupling was reported, 12 s tracks of about 830 frames bound the error from above, $\sigma \leq 0.40\ \mu\text{m}$, through a Cauchy–Schwarz constraint on held-out track halves. At 45–48 s sampling, noise then explains at most 13–23% of the coupling, so this coupling is robust to localization error. The standard calibration, the intercept of the mean-squared displacement, fails for cells: fast genuine motion masquerades as noise and leads to over-correction. The results argue for calibrating localization error, from fixed objects or long tracks, with any speed–persistence analysis.

I. INTRODUCTION

Maiuri *et al.* reported that across many cell types, faster cells turn less, and they traced this “universal coupling between cell speed and cell persistence” to actin flows [1]. Since then, speed–persistence and speed–turning correlations have been read as cell-intrinsic programs. Ganusov, Zenkov and Majumder showed that such correlations arise naturally when tracks are sampled at intervals comparable to the persistence time and persistence varies between cells [2]. This is a measurement effect requiring no biological coupling.

A second measurement effect works at the opposite end, at frame intervals that are short or displacements per frame that are small. Every tracked position carries a localization error. In single-particle tracking its effect on mean-squared displacements is well understood [3, 4]. Here we work out its effect on persistence. The mechanism is simple. Consecutive steps share a position, so independent position errors anticorrelate them. The anticorrelation matters most for cells that move little between frames. Speed heterogeneity alone then produces “faster cells are straighter”. We give the closed form, a noise-correctable test, and its results on lymph-node, in vitro and zebrafish T-cell tracks. The zebrafish data are those in which a speed–turning coupling was reported in vivo [5] and which motivated Ref. [2]. We make no claim about the data of Ref. [1].

II. THEORY

Take a two-dimensional persistent random walk whose velocity is an Ornstein–Uhlenbeck (OU) process. Each coordinate has velocity variance v^2 and persistence time P . Positions are sampled every Δt , and each coordinate is observed with independent Gaussian error of variance σ^2 . With $x = \Delta t/P$, the true steps of one coordinate have variance and lag-1 covariance

$$V = 2v^2P^2(x - 1 + e^{-x}), \quad (1)$$

$$C_1 = v^2P^2(1 - e^{-x})^2. \quad (2)$$

The error adds $2\sigma^2$ to the step variance. Since consecutive steps share one position, it also adds $-\sigma^2$ to their covariance. The measured consecutive steps are therefore isotropic Gaussian vectors with correlation

$$\rho = \frac{C_1 - \sigma^2}{V + 2\sigma^2}, \quad (3)$$

and the mean cosine of the turning angle between them is

$$E[\cos \theta] = \frac{\pi}{4} \rho {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 2; \rho^2\right). \quad (4)$$

The mean measured speed is $\sqrt{\pi(V + 2\sigma^2)/2}/\Delta t$. Equations (3) and (4) agree with Monte Carlo simulation to within 1–2% over five parameter sets.

ρ depends on speed only through $\sigma^2/(v^2P^2)$, the noise relative to the distance a cell covers in one persistence

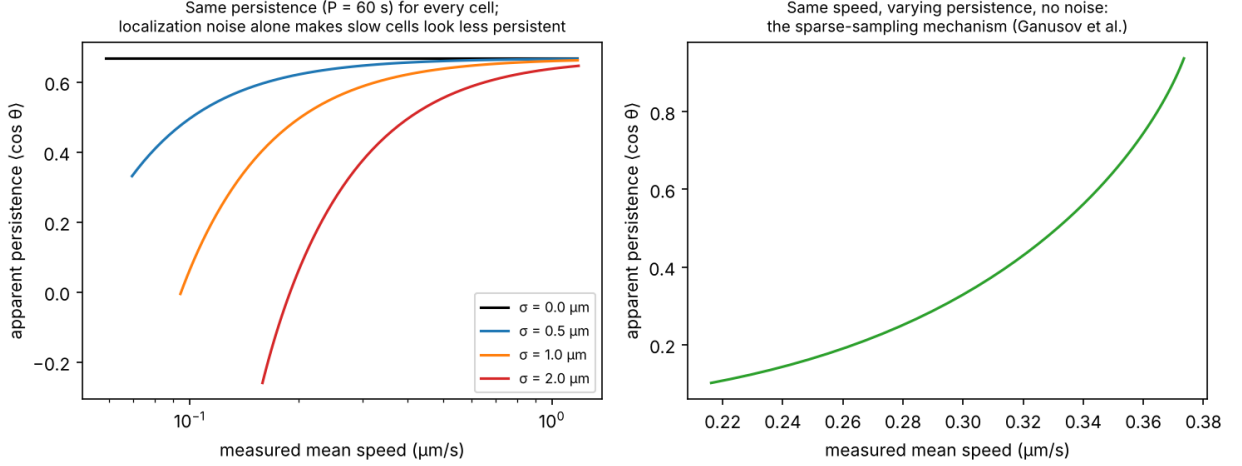


FIG. 1. Two routes to an artefactual speed–persistence coupling, from the closed form of Eq. (4). Left: every cell has the same persistence time $P = 60$ s, frames are every $\Delta t = 24$ s, and cells differ only in speed. Localization error of size σ makes slow cells appear less persistent. Right: equal speeds, varying persistence and no noise, which is the sparse-sampling mechanism of Ref. [2].

time. At fixed P it therefore rises monotonically with v (Fig. 1, left). Any spread of speeds in the population then shows up as a positive speed–persistence correlation. With realistic parameters ($P = 60$ s, $\Delta t = 24$ s, speeds $0.1\text{--}1\text{ }\mu\text{m/s}$), $\sigma = 0.5\text{ }\mu\text{m}$ halves the apparent persistence of cells moving at $\sim 0.07\text{ }\mu\text{m/s}$. Sparse sampling works the other way (Fig. 1, right): noise biases fast frame rates, while sparse sampling biases slow ones.

Inversion. For each cell c we use the lag-1 step correlation $r_c = \hat{C}_{1,c}/\hat{V}_c$, where the uncentred moments are pooled over both coordinates. Equation (3) inverts exactly,

$$r_c^* = \frac{\hat{C}_{1,c} + \sigma^2}{\hat{V}_c - 2\sigma^2}, \quad (5)$$

which removes the noise bias whatever the cell’s speed or persistence. The turning-angle cosine of Eq. (4) has no such simple inverse, so it is reported as a secondary statistic only.

III. METHOD

Effect size. We split cells into terciles of measured speed. The persistence gap G is the mean r of the fastest third minus that of the slowest third. With $G_{\text{corr}}(\sigma)$ the same gap computed from r^* , the fraction of the gap attributable to noise of size σ is

$$f_{\text{noise}}(\sigma) = 1 - G_{\text{corr}}(\sigma)/G_{\text{raw}}. \quad (6)$$

Cells with $\hat{V}_c - 2\sigma^2 < 0.2\hat{V}_c$, which have too little signal to correct, are dropped from both gaps, and their number is reported. Uncertainties come from a 300-replicate bootstrap over cells.

Why a curve. On synthetic populations, the correction with the true σ behaves as intended. With no coupling and $\sigma = 1\text{ }\mu\text{m}$ it attributes 98% of the gap to noise. With a genuine coupling (persistence proportional to speed) and $\sigma = 0.2\text{ }\mu\text{m}$ it attributes 18%. But σ itself is poorly identified from tracks of 6–40 positions. A lag-covariance estimator that is unbiased under homogeneous persistence ranged from 0.48 to $1.24\text{ }\mu\text{m}$ over ten seeds for a true value of 1.0 , and it fails when persistence varies between cells. The primary output is therefore the curve $f_{\text{noise}}(\sigma)$. It is summarized by $\sigma_{1/2}$, the smallest σ at which noise explains half the gap.

Preregistration. The protocol was committed to a public repository before any speed or persistence statistic of the real tracks was computed. Four dated amendments, made on synthetic data before the real-data run, record how the method arrived at this form. For the in vitro data the protocol was rerun unchanged, apart from the frame interval, together with fixed decision rules. Coupling is called *not robust to localization noise* if $\sigma_{1/2}$ is at most one pixel, and *robust* if $\sigma_{1/2}$ exceeds two pixels. Synthetic self-tests at each frame interval had to pass first, and did.

Data. (i) Two-photon tracks of T cells, B cells and neutrophils in mouse lymph nodes, distributed with cell-trackR [6]: 2D, $\Delta t = 24$ s, with 199, 74 and 363 cells of at least 6 positions. (ii) T cells migrating in vitro on ICAM-1 or VCAM-1, imaged by $10\times$ phase contrast every 30 s for 10 min and tracked with StarDist and TrackMate [7, 8]. We used the CellTracksColab compilation [9, 10], with pixel size $0.645\text{ }\mu\text{m}$. It gives 820 (ICAM-1) and 1083 (VCAM-1) gap-free tracks of at least 6 positions; 122 tracks with missing frames were excluded. (iii) T cells in larval zebrafish [5]: 712 control and 236 rock-out tracks at $\Delta t = 45$ s, and 29 tracks at $\Delta t = 12$ s with

TABLE I. Speed–persistence gap and its noise sensitivity. G_{raw} is the gap in the noise-correctable measure r . ρ_S is the Spearman correlation of speed with mean turning cosine, the conventional metric; $\rho_S^{(2)}$ is the same after keeping every second frame. f_{noise} is evaluated at $\sigma = 0.3 \mu\text{m}$. LN: in vivo lymph node. Bootstrap 16–84% ranges of $\sigma_{1/2}$: B cells 0.3–0.4, ICAM-1 0.5–0.6, VCAM-1 $0.4 \mu\text{m}$.

Dataset	Cells	G_{raw}	ρ_S	$\rho_S^{(2)}$	f_{noise}	$\sigma_{1/2} (\mu\text{m})$
T cells (LN)	199	−0.04	0.28	0.39	–	none
B cells (LN)	74	0.21	0.61	0.69	0.48	0.4
Neutrophils (LN)	363	−0.11	0.02	0.25	–	none
ICAM-1 T cells	820	0.93	0.83	0.80	0.16	0.6
VCAM-1 T cells	1083	0.73	0.76	0.72	0.42	0.4

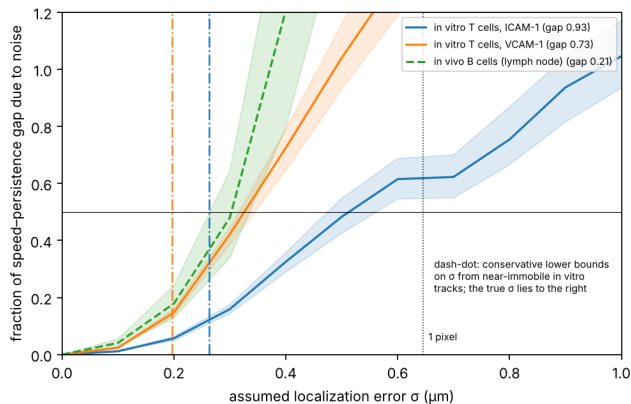


FIG. 2. Fraction of the speed–persistence gap explained by localization error of size σ , with 16–84% bootstrap bands. Dash-dot lines are the conservative lower bounds on σ for the in vitro pipeline, from nearly immobile tracks (Sec. V). The dotted line marks one camera pixel.

about 830 frames each. For each track we use its longest gap-free segment.

IV. RESULTS

Table I and Fig. 2 summarize the results.

In vivo. In the noise-correctable measure, T cells and neutrophils show no positive gap, so there is no coupling for noise to explain. B cells do show one, but a localization error of only $0.3\text{--}0.4 \mu\text{m}$ would explain between half and all of it. That is plausible for centroids of $\sim 7 \mu\text{m}$ cells in two-photon stacks, but it is unmeasured. The conventional metric disagrees with the noise-correctable one in the direction the theory predicts. For T cells, mean turning cosine rises with speed ($\rho_S = 0.28$; cosine gap 0.20) even though G_{raw} is slightly negative. Halving the frame rate raises ρ_S in all three datasets, consistent with the sparse-sampling mechanism of Ref. [2]. For these data, then, noise and sampling bias the conventional statistic

in the same direction, from opposite ends of the frame-interval range.

In vitro. The coupling is strong: $G_{\text{raw}} = 0.73\text{--}0.93$ and $\rho_S = 0.76\text{--}0.83$. Halving the frame rate slightly lowers it, so sparse sampling is not driving it. Still, $\sigma_{1/2}$ is $0.6 \mu\text{m}$ on ICAM-1 and $0.4 \mu\text{m}$ on VCAM-1, both below the $0.645 \mu\text{m}$ pixel. The frozen verdict for both conditions is therefore *not robust to localization noise*. This means the tracks alone cannot separate the coupling from a noise artefact; it does not mean the coupling is an artefact. At $\sigma = 0.2 \mu\text{m}$, about a third of a pixel and plausible for StarDist centroids of $\sim 10 \mu\text{m}$ cells, noise explains only 6–15%.

V. BOUNDING THE NOISE FROM THE TRACKS

This analysis was done after the frozen verdict. For any subset of tracks, persistent motion contributes $C_1 \geq 0$, so $\sigma^2 \geq -\langle \hat{C}_1 \rangle$ over that subset. The slowest 5% of in vitro tracks, ranked by $\hat{V}$, have mean lag-1 step correlation -0.38 (ICAM-1) and -0.26 (VCAM-1). Pure noise gives -0.5 . They give $\sigma \geq 0.26 \mu\text{m}$ (16–84%: $0.25\text{--}0.28$) and $\sigma \geq 0.20 \mu\text{m}$ ($0.19\text{--}0.21$), so $f_{\text{noise}} \geq 0.12$ and ≥ 0.14 . The bound rises with the selected fraction: for ICAM-1 it is $0.23, 0.26, 0.29$ and $0.30 \mu\text{m}$ at 2, 5, 10 and 20%. That is the pattern expected if centroid error grows with cell deformation.

Selecting tracks on $\hat{V}$ and then evaluating them on the same $\hat{V}$ biases both moments low, because it picks the realizations whose noise happened to be small. For the lower bound the bias is in the safe direction. In synthetic controls with track lengths taken from the data, immobile fractions of 3–30%, and a true $\sigma = 0.25 \mu\text{m}$, it returned $0.18\text{--}0.20 \mu\text{m}$. The tempting upper bound $\sigma \leq (\langle \hat{V} \rangle / 2)^{1/2}$ from the same selection, however, is *invalid*: in the same controls it returned $0.26, 0.21$ and $0.19 \mu\text{m}$, below the true value in two of three cases. Selecting on the first half of each track and evaluating on the disjoint second half removes the bias: in all three synthetic controls the resulting bracket contains the true $0.25 \mu\text{m}$. On the real tracks, though, it gives only loose upper bounds: $0.50 \mu\text{m}$ for VCAM-1 and $1.2 \mu\text{m}$ for ICAM-1. Cells that are slow in one half do not stay slow, because cells pause and resume. The tighter bound of Sec. VI is also loose for these 21-frame tracks ($0.5\text{--}0.8 \mu\text{m}$). The tracks therefore show that noise explains at least about an eighth of the in vitro coupling, but they cannot exclude that it explains most of it.

VI. CALIBRATING THE NOISE FROM LONG TRACKS

Tracks hundreds of frames long should make σ measurable. We tested this on the zebrafish data with a protocol frozen before the real-data run. It fitted each long

TABLE II. Zebrafish T cells [5]: localization-error bounds and the fraction of the gap attributable to noise. $\sigma_{\max}$ is the Cauchy–Schwarz split-half upper bound of Eq. (7), with its 16–84% bootstrap range, and f_{noise} is evaluated at it. The 45 s control movies use the bound from the 12 s movies (transferred). “12 s, coarsened” is the 12 s cells sampled every fourth frame (48 s).

Group	Cells	G_{raw}	$\sigma_{\max}$ (μm)	f_{noise}
12 s	29	0.44	0.40 (0.37–0.46)	1.8
12 s, coarsened	29	0.49	0.40 (same cells)	≤ 0.23
45 s control	711	0.58	0.40 (transferred)	≤ 0.13
45 s knockout	236	0.52	0.71 (0.66–0.78)	≤ 0.76

track’s per-coordinate mean-squared displacement at lags 1–10 frames with the OU form plus a white-noise intercept $2\sigma^2$, the standard approach in particle tracking [3]. The estimator passed synthetic self-tests that matched the real track lengths and had within-track switching of persistence. On the 12 s fish tracks it gave $\sigma = 0.49 \mu\text{m}$. Applied to the very tracks it was calibrated on, however, that value attributes 3.5 times the observed gap to noise. That is impossible, and it signals that the calibration is wrong. A *Dictyostelium* set compiled in the same repository (5 s frames) over-corrected even more.

The contradiction is direct. Because the true step covariance obeys $C_1 \leq V$ (Cauchy–Schwarz), white error must satisfy

$$\sigma^2 \leq \frac{1}{3} (\hat{V} - \hat{C}_1). \quad (7)$$

To avoid the selection bias of Sec. V, we rank tracks by $\hat{V}$ on their first halves and evaluate Eq. (7) on the slowest third of the disjoint second halves. The slowest 12 s fish tracks have lag-1 step correlation +0.20. White error of $0.49 \mu\text{m}$ would force about -0.5 .

The mechanism is a fast component of genuine motion that decorrelates within about a frame. It raises the short-lag MSD but leaves no negative lag-1 dip, so a single-persistence fit absorbs it into the noise intercept. We confirmed this on synthetic tracks with the real 12 s lengths, a fast velocity component carrying 0–60% of the variance, and true σ of 0.15, 0.3 and $0.5 \mu\text{m}$. The MSD intercept was biased up, for example from 0.15 to $0.29 \mu\text{m}$. The split-half bound of Eq. (7) stayed at or above the true value in all nine cases. The persistence correction of Eq. (5) needs only the white part, and Eq. (7) bounds exactly that part.

On the 12 s fish tracks the bound is $\sigma \leq 0.40 \mu\text{m}$ (Table II), which excludes the MSD calibration. Sampled at 12 s, even this bound would allow noise to explain the whole gap. Sampled every fourth frame (48 s), the same cells keep their coupling (G rises only from 0.44 to 0.49), and noise explains at most 23% of it, or 30% at the bootstrap 84% bound. Transferring the bound to the 45 s control movies, which assumes similar imaging settings, gives at most 13%. The coupling reported in Ref. [5]

is therefore robust to localization error at the authors’ sampling interval. This agrees with their own check at an assumed noise level, and here it rests on a calibration from the data. The slower knockout cells are more sensitive: noise explains up to three quarters of their gap within their own bound. These estimators were developed after the frozen run, so this section is post hoc.

VII. DISCUSSION

Localization error biases persistence in a speed-dependent way. It can therefore manufacture, inflate or mask a speed–persistence coupling. The bias is set by σ relative to the per-frame displacement, so it matters most for slow cells, short frame intervals and coarse pixels. It is complementary to the sparse-sampling bias [2], and the two act at opposite ends of the frame-interval range. For some datasets there may be no frame interval free of both.

Three practical points follow. First, persistence measured by the lag-1 step correlation can be corrected exactly through Eq. (5), whereas turning-angle statistics cannot. Second, σ is not reliably identifiable from short tracks, and the usual MSD-intercept calibration is biased upward for cells because their fast genuine motion counts as noise. The quantity that matters is the white part, which is bounded by the lag-1 structure through Eq. (7), evaluated on held-out track halves. Selection on noisy quantities can silently invalidate a bound, and a self-test can pass on synthetic data that lack a feature of real cells. Both happened here. Third, the clean remedy is experimental: image fixed cells or beads with the same pipeline, or record a subset of long, fast-sampled tracks as in Ref. [5], and report σ with the speed–persistence analysis. With such a calibration the curves of Fig. 2 turn directly into an attribution.

The OU walk is a null model, not a description of lymphocyte motility. The data are 2D, and the in vivo tracks are projections of 3D motion. Our persistence measure differs from that of Ref. [1]. The in vitro tracks are 21 frames long, and the zebrafish bound rests on 29 long tracks. None of the analysed datasets is from Ref. [1], and we draw no conclusion about those experiments.

DATA AND CODE AVAILABILITY

All analysis code, frozen protocols with their amendment history, self-test outputs and result files are available from the author (analysis scripts labelled labs 48–54). The track data are public [5, 6, 9] and are not redistributed; their checksums are recorded with the results.

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- [1] P. Maiuri *et al.*, Cell **161**, 374 (2015), doi:10.1016/j.cell.2015.01.056.
 - [2] V. V. Ganusov, V. S. Zenkov, and B. Majumder, Phys. Biol. **20**, 025001 (2023), doi:10.1088/1478-3975/acb18c.
 - [3] X. Michalet, Phys. Rev. E **82**, 041914 (2010), doi:10.1103/PhysRevE.82.041914.
 - [4] A. J. Berglund, Phys. Rev. E **82**, 011917 (2010), doi:10.1103/PhysRevE.82.011917.
 - [5] E. R. Jerison and S. R. Quake, eLife **9**, e53933 (2020), doi:10.7554/eLife.53933.
 - [6] I. M. N. Wortel *et al.*, ImmunoInformatics **1–2**, 100003 (2021), doi:10.1016/j.immuno.2021.100003.
 - [7] E. Fazeli *et al.*, F1000Research **9**, 1279 (2020), doi:10.12688/f1000research.27019.1.
 - [8] N. H. Roy and G. Jacquemet, “Combining StarDist and TrackMate example 2 – T cell dataset,” Zenodo (2020), doi:10.5281/zenodo.4034929.
 - [9] G. Jacquemet, “T cell dataset for CellTracksColab – 2,” Zenodo (2023), doi:10.5281/zenodo.8420011.
 - [10] E. Gómez-de-Mariscal *et al.*, PLoS Biol. **22**, e3002740 (2024), doi:10.1371/journal.pbio.3002740.